Chapter -7

← Loops in Python →

Sometimes we want to repeat a set of statements in our program. For instance : Print 1 to 1000.

Loops make it easy for a programmer to tell the computer, which set of instructions to repeat and how!

Types of loops in Python: Primarily these are two types of loops in Python.

1> While loop
2> For loop

We will look into these one by one.

While loop : The syntax of a while loop looks like this:

```
While (Condition):
    # "Body of the loop"
```

⇒ The block keep executing until the condition is true!

In while loops, the condition is checked first. If it evaluats to True, the Body of the loops is executed, otherwise not!

If the loops is entered, the process of [condition check & execution] is continued untill the condition becames False

Quick Quiz: Write a python to print 1 to 50 using a while (~~a~~ x ~~w~~) loop.

An Example -

```
i = 0
While(i<5):
    Print ("Harry"):
    i = i+1
```

⇒ Prints "Harry" - 5 times!

Note: If The condition never becomes False, the loop keeps getting executed.

Quick Quiz: Write a program to print the content of a list using while loops.

For loop → A for loop is used to itrate through a sequence like, tuple or string.

The syntax of a for loop looks like this:

```
l = [1, 7, 8]
For item in l:        => Print 1, 7, and 8
    Print(item)
```

Range function in Python - The range function in Python is used to generate a sequence of numbers. We can also specify the start, stop and step-size as follow:

```
range(start, stop, step_size)
```

↳ step size is usually not used with range()

An Example demonstrating rang() function.

```
For i in rang(0, 7):   → rang(7) can also be used
    Print(i)           → Prints 0 to 6
```

For loop with else - An optional else can be used with a for loop if the code is to be executed when the loop exhausts. Example:

```
l = [1, 7, 8]
For item in l:
    Print(item)
else:
    Print("Done")  → This is print
```

Output: 1
7
8
Done

The Break statement - 'break' is used to come out of the loop when encountered. It instructs the program to - Exit the loop now.

Example:

```
for i in range(0,80):
    Print(i)
    if i== 3:
        break
```

⇒ This will print 0, 1, 2 and 3.

The continue statement - 'continue' is used to stop the current itration of the loop and continue with the next one. It instructs the program to "Skip this itration".

Example:

```
For i in range(4):
    Print("Printing")
    if i ==2:
        Continue
    Print(i)
```

Pass statement - Pass is a null statement in Python. It instructs to "Do nothing"

Eample:

```
l =[1,7,8]
For item in l:
    Pass
```

⇒ Without Pass, the Program will throw an error.

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EXPLORER

CHAPTER-7_LOOPS.PY

- break_and_continue.py
- for.py
- pass.py
- practice_set.ipynb
- while_loop.py

OUTLINE

TIMELINE

for.py > list1

```python
list1=[1,"virendra",False,"this","shivaji",2

for item in list1:
    print(item)

for item in range(len(list1)):
    print(list1[item])

#for loop with else
for item in list1:
    print(item)
else:
    print("Loop is over")
```

while_loop.py > ...

```python
# while loop
i=1
while i<6:
    print(i)
    i+=1
'''
Output:
1
2
3
4
5
'''
list1=[1,"virendra",False,"this","shivaji",2

for item in list1:
    print(item)

j=0
while j<len(list1):
    print(list1[j])
    j+=1

for item in range(len(list1)):
    print(list1[item])
```

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EXPLORER

CHAPTER-7_LOOPS.PY

- 1.png
- break_and_continue.py
- for.py
- pass.py
- practice_set.ipynb
- while_loop.py

OUTLINE

TIMELINE

break_and_continue.py > ...

```python
for i in range(1,22):
    if i ==20:
        break # it will break the loop when i is 2
    print(i)

for i in range(1,22):
    if i ==20:
        continue # skip this itration
    print(i)
```

pass.py > ...

```python
for i in range(1,6):
    pass # it will do nothing and move to ne
for i in range(1,10):
    if i<=8:
        continue
    else:
        print(i)
```

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